# In-company or Research Internship MSc Computing Science – Starting form

*To be filled in and signed by the student and the supervisor(s) at the beginning of the internship.*

Course code: WMCS021-15
Credits (EC): 15 points (420 hours)

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## 1. General information

Student name and Signature: Wesley Haverkort

Student number: s3811808

Project title: Semantic Scene Understanding on Gaussian Splats

Date: 25/03/2026

Starting meeting: 13 April 2026

Final assessment: 22 Juni 2026

First supervisor (name + signature): dr. Kailai Li

Second supervisor (name + signature): dr. Jiapan Guo

Daily supervisor (if applicable): Yixuan Qiu

External supervisor (if applicable):

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## 2. Project description:

This internship project focuses on extending an existing RGB-D perception pipeline towards full scene understanding. The current system uses a ROS2-based architecture in which objects are detected using YOLO, segmented at the instance level, and projected into a 3D point cloud using depth information. This approach is object-centric and only processes selected regions of the image.

The goal of this project is to transition from this object-centric pipeline to a semantic segmentation approach that assigns a class label to every pixel in the full image. This allows the system to represent and reason about the entire scene instead of individual objects. As a baseline, the system will perform semantic segmentation over the full image without requiring user interaction. The existing click-based

interface will be retained as an optional feature to select specific semantic regions of interest.

The semantic segmentation output will be converted into binary masks and projected into 3D using the existing depth processing pipeline. This results in semantically labeled point clouds that represent the scene in a more complete and structured way and serve as a baseline 3D representation.

In a second phase, the project will explore the integration of Gaussian Splatting, a recent technique for efficient and continuous 3D scene representation, into the pipeline. As an initial step, existing Gaussian Splatting methods from the literature will be used to reconstruct a 3D scene representation from multi-view RGB observations, without incorporating semantic information. This serves as a baseline to isolate the effect of multi-view 3D reconstruction.

Recent work has demonstrated the potential of combining Gaussian Splatting with semantic segmentation for multi-view scene understanding (Qi, Chen, Zhang, Zou, & Shi, 2025)

After establishing this baseline, the project will investigate more advanced approaches by incorporating semantic information into the Gaussian representation. This includes associating semantic labels with Gaussians and exploring strategies to integrate semantically relevant regions selectively. This enables combining observations from multiple viewpoints into a unified and semantically-aware 3D representation.

The project will evaluate multiple configurations: standard frame-by-frame semantic segmentation, semantic segmentation combined with point cloud representations, Gaussian Splatting without semantic information, and semantically-aware Gaussian Splatting. The evaluation will consider different scene types, including indoor and outdoor environments, as well as static and dynamic scenes, to ensure that the comparison reflects realistic perception challenges. This allows isolating the effects of multi-view fusion and semantic integration.

The main research objective is to evaluate whether Gaussian Splatting improves semantic consistency, spatial coherence, and robustness to noise compared to standard frame-by-frame semantic segmentation and point cloud-based representations.

The system will be implemented within a modular ROS2 framework and will reuse existing components for RGB-D input, visualization, and point cloud generation. Development will focus on integrating semantic segmentation, adapting the interaction model, and extending the system with Gaussian Splatting. Pretrained

semantic segmentation models will be used as a baseline, with optional fine-tuning if required for improved performance in the target environment.
**RQ:** To what extent can Gaussian Splatting improve semantic consistency, spatial coherence, and robustness in 3D scene representations compared to standard frame-by-frame semantic segmentation and point cloud-based methods?

## 3. Methodology and timeline:

**Methodology**

The project will be implemented within a ROS2-based perception framework using RGB-D data as input. The existing system will be extended in a modular way, ensuring that core components such as data acquisition, visualization, and 3D projection remain unchanged.

First, a semantic segmentation model will be integrated into the pipeline to generate per-pixel class labels from RGB input. Instead of committing to a single architecture upfront, multiple candidate models will be considered, including transformer-based methods such as SegFormer (Xie, 2021) and more recent SAM-based approaches. These models differ in terms of segmentation performance, generalization ability, and computational requirements. Therefore, an initial comparison will be conducted, and the final model will be selected based on performance, robustness, and real-time feasibility within the target system.

A pretrained deep learning model (e.g., implemented in PyTorch) will be applied to full RGB images to generate per-pixel class labels. This replaces the current object-centric instance segmentation approach. The system will initially operate on full-image semantic segmentation without requiring user interaction.

Second, the semantic segmentation output will be converted into binary masks and combined with depth data to generate semantically labeled 3D point clouds. This step reuses the existing projection pipeline and serves as a baseline 3D representation of the scene.

Third, the existing click-based interaction will be retained as an optional feature. Instead of selecting an object, a user-selected pixel will determine the semantic class of interest. Optionally, a connected region around the selected pixel can be extracted to refine the selected region.

In a later phase, Gaussian Splatting will be integrated to construct a continuous 3D scene representation. As an initial step, a state-of-the-art semantic Gaussian Splatting method from the literature will be selected as the baseline for 3D scene representation. This allows the system to directly incorporate semantic information into the Gaussian-based model and provides a more meaningful reference point for evaluation.

After establishing this baseline, a state-of-the-art semantic Gaussian Splatting method will be used to construct a semantically-aware 3D representation. The focus of this project is not on modifying the underlying method, but on evaluating its performance in comparison to frame-by-frame semantic segmentation and point cloud-based representations. This enables combining observations from multiple viewpoints into a unified 3D representation.

Experiments will initially be conducted using recorded RGB-D sequences collected in a controlled indoor lab environment to enable reproducible evaluation across multiple viewpoints. These recordings will include both static and dynamic scenes, as well as varying levels of occlusion and object interaction. If time permits, a near-real-time version of the pipeline will be explored.

Additionally, outdoor recordings may be captured using a mobile device (e.g., iPhone) to evaluate generalization beyond the lab environment. Since not all methods require or support depth information, this setting will be used as an exploratory evaluation and will focus on methods that can operate on RGB input only, unless depth data is available.

Finally, the system will be evaluated by comparing:
- instance-based vs semantic segmentation
- point cloud vs Gaussian Splatting representations

The system will also be evaluated under controlled perturbations, such as motion blur, noisy depth, missing depth regions, and partial occlusions, to assess robustness.

The evaluation will include both qualitative and quantitative analysis. For semantic segmentation performance, standard metrics such as precision, recall, Intersection over Union (IoU), and Dice coefficient will be used. For evaluating the quality of the 3D reconstruction and rendered views from Gaussian Splatting, image-based metrics such as Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR) and Learned Perceptual Image Patch Similarity (LPIPS) will be applied.

These metrics will be used to assess semantic consistency, spatial coherence, robustness to noise, and stability across multiple viewpoints.

The project will primarily use Python, ROS2, PyTorch, and visualization tools such as RViz.

**Timeline and milestones (10 weeks)**

Week 1–2: Literature study and system design

A focused study will be carried out on semantic segmentation methods and Gaussian Splatting, with particular attention to how these can be integrated into the existing ROS2-based RGB-D pipeline. The current system architecture will be reviewed from an

integration perspective, and the technical approach for extending it will be defined.
Milestone: Clear integration plan for semantic segmentation and Gaussian Splatting.

Week 2–3: Semantic segmentation integration
A semantic segmentation model will be integrated into the pipeline. The current crop-based instance segmentation will be replaced by full-image semantic segmentation, and semantic masks will be generated and verified.
Milestone: Working full-image semantic segmentation pipeline.

Week 4–5: Interaction and 3D projection
The click-based interaction will be adapted to semantic selection. Semantic regions will be converted into binary masks and combined with depth data to generate 3D point clouds.
Milestone: End-to-end semantic RGB-D to 3D point cloud pipeline.

Week 5–8: Gaussian Splatting integration
Gaussian Splatting will be integrated into the existing pipeline based on the design choices from the earlier research phase. Semantic labels will be associated with the Gaussian-based representation to enable multi-view fusion and scene-level reasoning.
Milestone: Initial semantically-aware Gaussian-based 3D representation.

Week 9: Evaluation and comparison
The system will be evaluated by comparing semantic and instance-based approaches, as well as point cloud and Gaussian-based representations. Results will be analyzed in terms of semantic consistency, robustness, and practical limitations.
Milestone: Completed experiments and evaluation results.

Week 10: Finalization
The implementation will be optimized and cleaned up. The final report and presentation will be prepared, and code and results will be documented.
Milestone: Final report and deliverables completed.

## 4. Deliverables:

Source code,
Documentation on the code,
Experimental results and datasets,
Evaluation and analysis,
Presentation.

## 5. Grading

Scientific quality of research and/or technical contribution: 40%
Project management and interpersonal skills: 20%
Final Presentation: 20%
Report/Thesis: 20%

## Works Cited

Qi, Z., Chen, H., Zhang, H., Zou, Z., & Shi, Z. (2025). Efficient Semantic Splatting for Remote Sensing Multiview Segmentation. *IEEE Transactions on Geoscience and Remote Sensing*.

Xie, E. W. (2021). SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers. *arXiv*.

Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3d gaussian splatting for real-time radiance field rendering. ACM Trans. Graph., 42(4), 139-1.

Qin, M., Li, W., Zhou, J., Wang, H., & Pfister, H. (2024). Langsplat: 3d language gaussian splatting. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 20051-20060).

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Feng, G., Chen, S., Fu, R., Liao, Z., Wang, Y., Liu, T., ... & Dai, B. (2025). Flashgs: Efficient 3d gaussian splatting for large-scale and high-resolution rendering. In Proceedings of the Computer Vision and Pattern Recognition Conference (pp. 26652-26662).